

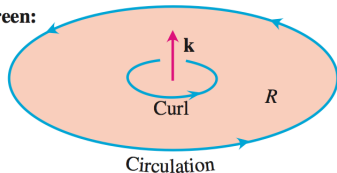
Topic: Stokes' Theorem

I. Stokes' Theorem

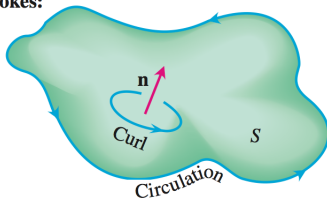
Thm: Let S be a surface with boundary C and $\mathbf{F}(x, y, z)$ be a vector-valued function defined on S . Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} =$$

Green:



Stokes:



Note that Green's Theorem is a special case of Stokes' theorem when S is a plane region R . In this case, $\mathbf{n}$ is just _____ and _____

$$\oint_C \mathbf{F} \cdot d\mathbf{r} =$$

Ex: $\mathbf{F} = [-5y, 4x, z]$, and C is the circle $x^2 + y^2 = 16, z = 4$. Find $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Note: $\text{curl } \mathbf{F} =$ and $\mathbf{n} =$

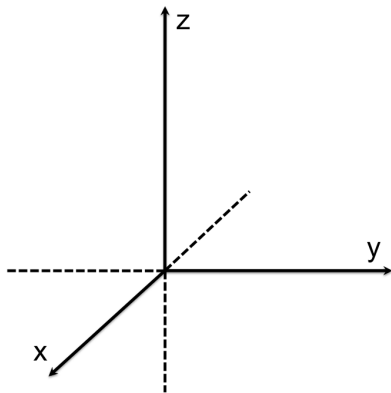
So

$$\oint_C \mathbf{F} \cdot d\mathbf{r} =$$

Let $\mathbf{F} = [4yz, 6z - 3xy, x]$. Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for each of the following curves:

- a. C is the union of the straight paths from $(0, 0, 0)$ to $(0, 0, 5)$ to $(0, 3, 5)$ and back to $(0, 0, 0)$
- b. C is the union of the straight paths from $(0, 0, 5)$ to $(2, 0, 0)$ to $(0, 5, 0)$ and back to $(0, 0, 5)$

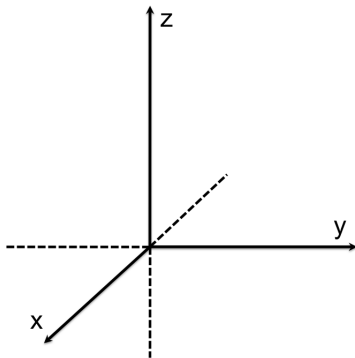
$$\oint_C \mathbf{F} \cdot d\mathbf{r} =$$



Need the Eqn of the plane which passes through: $(2, 0, 0), (0, 5, 0), (0, 0, 5)$

General Form: $ax + by + cz = d$

By observation of where the plane intersects with each axis, we can determine that the



$$\oint_C \mathbf{F} \cdot d\mathbf{r} =$$

Ex: $\mathbf{F} = [-y, 2z, 0]$, C is the boundary of side of the cylinder $y^2 + z^2 = 4$, with $z \geq 0$, $0 \leq x \leq 3$ and oriented outward. Graph the surface and use Stokes' thm to compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.

$$\text{curl } \mathbf{F} = \quad \& \quad \mathbf{n} =$$

Using coordinates

$$x =$$

$$y =$$

$$z =$$

$$\mathbf{r}_u =$$

$$\mathbf{r}_v =$$

$$\mathbf{N} =$$

Note:

$$\Rightarrow \mathbf{N} =$$

$$\Rightarrow \text{curl } \mathbf{F} \cdot \mathbf{N} =$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dA$$

